Incident Response Investigation Report

1. Incident Description

A simulated cybersecurity incident was conducted in an isolated lab environment to practice SOC analyst investigation procedures.

- Target system: Ubuntu 24.04 LTS, IP 192.168.56.102 (user: ligs, sudo-privileged)
- Attacker/Monitoring system: Kali Linux, IP 192.168.56.103
- Incident type: Simulated SSH brute-force attack followed by unauthorized command execution and deployment of a persistent outbound beacon (reverse-connection style)
- Detection method: Real-time log monitoring (auth.log), network connection analysis (ss -tunap), and process monitoring (ps aux)

2. Evidence Collected

File	Description
logs/failed_logins_confirmed.txt	SSH brute-force failed login entries
logs/accepted_logins.txt	Successful SSH login entries
logs/login_history.txt	Full login history (last -a)
network_analysis/suspicious_connections.txt	ss -tunap output showing active connections
network_analysis/capture.pcap	Packet capture of beacon traffic on port 4444
indicators/.update.sh	Recovered malicious script
indicators/file_hash.txt	SHA-256 hash of malicious script
indicators/process_list.txt	Process snapshot showing malicious nc/.update.sh
indicators/analysis_summary.txt	Analyst notes and findings

3. Indicators of Compromise (IoC)

Type	Value	Source
Attacker/C2 IP	192.168.56.103	auth.log, ss -tunap
C2 port	4444	ss -tunap, pcap
Domain	None observed (direct IP-based C2, no DNS used)	N/A
Malicious file	/tmp/.update.sh	find, ps
File hash (SHA-256)	950e6fb133397c8a49345418e51a2568ccb73bc8833c811d156f0bc5c86ecb8	sha256sum
Process	nc (child of .update.sh)	ps aux
Persistence	crontab @reboot entry	crontab -l

Type	Value	Source
Targeted/compromised account	ligs (sudo-privileged)	auth.log

4. Attack Timeline

Time (+05:30)	Event
2026-10-14 20:06:39 - 20:07:33	Repeated failed SSH login attempts against 'ligs' from 192.168.56.103 (7 attempts, automated pattern)
[2026-10-14 20:09:48 (+05:30)]	Successful login to 'ligs' account
Unknown (not logged)	Reconnaissance commands executed: whoami, id, cat /etc/passwd
2026-10-13 01:05:50	Malicious script /tmp/.update.sh created
[2026-10-14 20:46:56 (IST, converted from EDT capture)]	Outbound beacon connection established to 192.168.56.103:4444
[confirm if present]	removed during remediation

Note: Exact timestamp for reconnaissance command execution could not be determined, as auditd exec-level logging was not configured on the target at the time of the incident. This was identified as a monitoring gap and addressed during remediation (see Section 6).

5. Containment Actions

Action	Status
Block malicious IP (ufw deny from 192.168.56.103)	Executed
Kill malicious processes (pkill -f /tmp/.update.sh, manual kill of residual nc PID)	Executed
Remove cron persistence (crontab -r)	Executed
Disable compromised account (usermod -L ligs)	Documented, not executed — account also served as investigator's access credential for this lab exercise
Isolate system from network (ip link down)	Documented, not executed — would have terminated the active investigation session; targeted IP blocking and process termination achieved equivalent containment

6. Remediation Recommendations

- Password reset performed for account 'ligs'
- Malicious file /tmp/.update.sh removed and deletion confirmed
- System packages updated (apt update && apt upgrade)
- fail2ban installed and enabled to automatically block future brute-force attempts
- auditd exec-level logging enabled to close the command-execution visibility gap identified during this investigation
- Recommended (for production use): disable SSH password authentication in favor of key-based authentication, and disable direct root login

7. Lessons Learned

- Dynamic (DHCP) IP addressing on the investigator's own host created temporary ambiguity when correlating historical login events — real-time ARP/neighbor table capture during an active incident would prevent this kind of evidence aging out.
- Command-level audit logging (auditd exec rules) was not enabled by default, leaving a visibility gap for reconstructing attacker actions after initial access — now remediated.
- Multiple redundant instances of the malicious process were found running simultaneously, which should prompt a more thorough initial sweep for all related processes rather than assuming a single instance.

8. Appendix

- Evidence hash: 950e6fb133397c8a49345418e51cb18bcecb73bc8833c811d156f0bc5c86ecb8 (/tmp/.update.sh)
- Full command log available in indicators/analysis_summary.txt